Branny Branfield

ITVNA0-22

Formative Assessment

Bedfordview Campus: Eduvos

Student Number : EDUV12936584

Question 1



Question 2 (create two VLANs on managed switch:)

- VLAN 10:Staff
- VLAN 20:Students

```
Switch>en
Switch#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#vlan ?
  <1-4094>  ISL VLAN IDs 1-1005
Switch(config)#vlan 10
Switch(config-vlan)#name Staff
Switch(config-vlan)#do sh vlan br

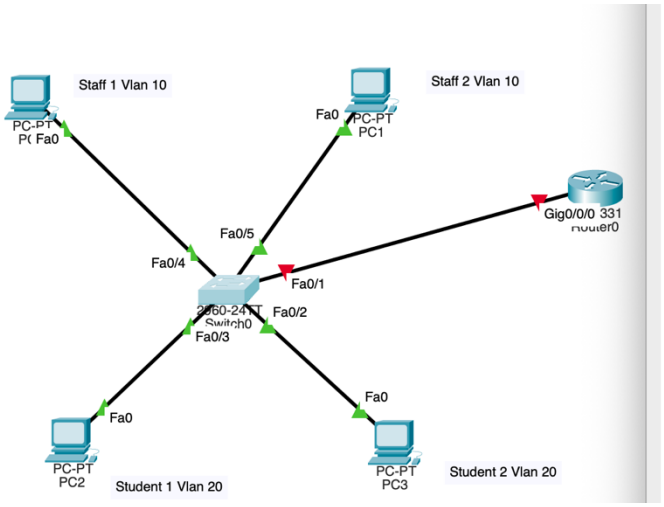
VLAN Name                              Status    Ports
---- --------------------------------- --------- -------------------------------
1    default                           active    Fa0/1, Fa0/2, Fa0/3, Fa0/4
                                                 Fa0/5, Fa0/6, Fa0/7, Fa0/8
                                                 Fa0/9, Fa0/10, Fa0/11, Fa0/12
                                                 Fa0/13, Fa0/14, Fa0/15, Fa0/16
                                                 Fa0/17, Fa0/18, Fa0/19, Fa0/20
                                                 Fa0/21, Fa0/22, Fa0/23, Fa0/24
                                                 Gig0/1, Gig0/2
10   Staff                             active
1002 fddi-default                      active
1003 token-ring-default                active
1004 fddinet-default                   active
1005 trnet-default                     active
Switch(config-vlan)#exit
Switch(config)#vlan 20
Switch(config-vlan)#name Students
Switch(config-vlan)#do sh vlan br

VLAN Name                              Status    Ports
---- --------------------------------- --------- -------------------------------
1    default                           active    Fa0/1, Fa0/2, Fa0/3, Fa0/4
                                                 Fa0/5, Fa0/6, Fa0/7, Fa0/8
                                                 Fa0/9, Fa0/10, Fa0/11, Fa0/12
                                                 Fa0/13, Fa0/14, Fa0/15, Fa0/16
                                                 Fa0/17, Fa0/18, Fa0/19, Fa0/20
                                                 Fa0/21, Fa0/22, Fa0/23, Fa0/24
                                                 Gig0/1, Gig0/2
10   Staff                             active
20   Students                          active
1002 fddi-default                      active
1003 token-ring-default                active
1004 fddinet-default                   active
1005 trnet-default                     active
Switch(config-vlan)#
```

Creating two VLANs “VLAN 10:Staff” and “VLAN 20:Students”

VLAN configuration on the Switch.

- Assign appropriate names to each VLAN for clarity
- Assign Client Machines to VLANs



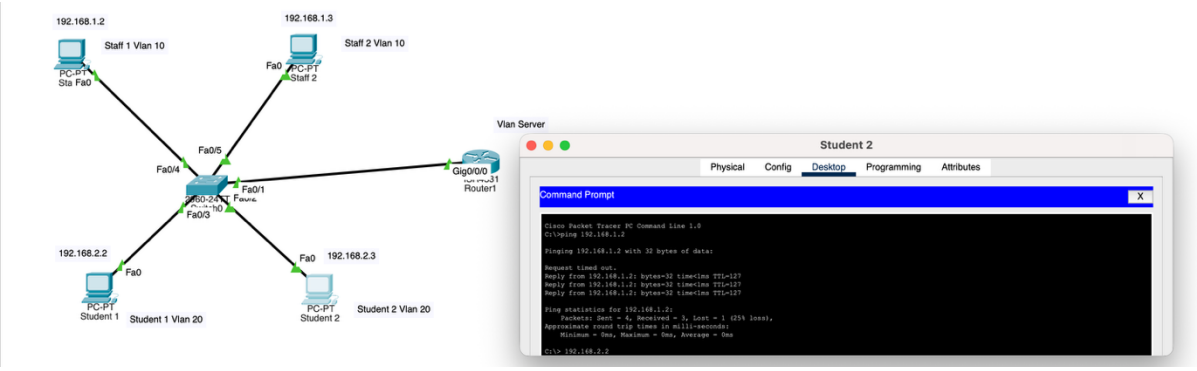

```
Switch#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#int fa0/3
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 20
Switch(config-if)#exit
Switch(config)#int fa0/2
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 20
Switch(config-if)#exit
Switch(config)#int fa0/4
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 10
Switch(config-if)#exit
Switch(config)#int fa0/5
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 10
Switch(config-if)#exit
Switch(config)#exit
Switch#
%SYS-5-CONFIG_I: Configured from console by console

Switch#sh vlan br

VLAN Name                              Status    Ports
---- --------------------------------- --------- -------------------------------
1    default                           active    Fa0/1, Fa0/6, Fa0/7, Fa0/8
                                                 Fa0/9, Fa0/10, Fa0/11, Fa0/12
                                                 Fa0/13, Fa0/14, Fa0/15, Fa0/16
                                                 Fa0/17, Fa0/18, Fa0/19, Fa0/20
                                                 Fa0/21, Fa0/22, Fa0/23, Fa0/24
                                                 Gig0/1, Gig0/2
10   Staff                             active    Fa0/4, Fa0/5
20   Students                          active    Fa0/2, Fa0/3
1002 fddi-default                      active
1003 token-ring-default                active
1004 fddinet-default                   active
1005 trnet-default                     active
Switch#
```

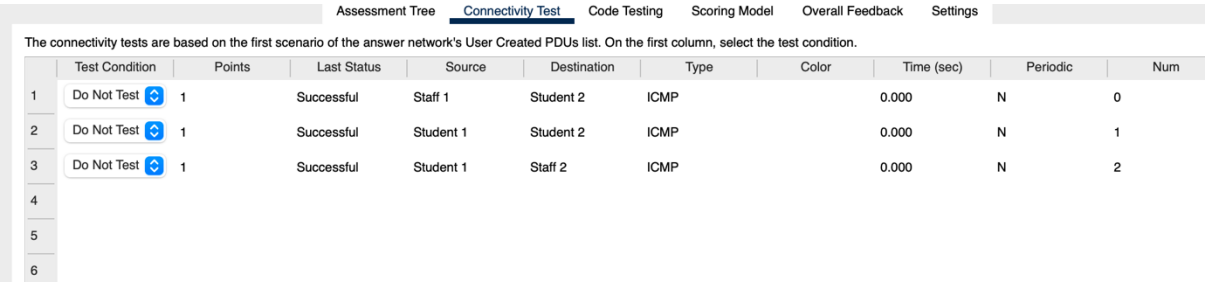
Assigning Client Machines to VLAN's with appropriate names "Student 1 and 2 to VLAN 20" and "Staff 1 and 2 to VLAN 10". 4 Clients to switch.

- Use ping tests to confirm that devices in different VLANs cannot communicate directly without routing.



Used Ping and confirmed that devices in different VLANs cannot communicate directly without routing. "Request timed out."
"Student 2 to Staff 1 " .

- Results of connectivity tests



Assessment Tree | Connectivity Test | Code Testing | Scoring Model | Overall Feedback | Settings

The connectivity tests are based on the first scenario of the answer network's User Created PDUs list. On the first column, select the test condition.

	Test Condition	Points	Last Status	Source	Destination	Type	Color	Time (sec)	Periodic	Num
1	Do Not Test	1	Successful	Staff 1	Student 2	ICMP		0.000	N	0
2	Do Not Test	1	Successful	Student 1	Student 2	ICMP		0.000	N	1
3	Do Not Test	1	Successful	Student 1	Staff 2	ICMP		0.000	N	2
4										
5										
6										

Results of connectivity between Client Devices in the same VLANs and in different VLANs.

Virtual Local Area Network (VLAN) Improves campus performance by dividing single networks into multiple isolated logical segments restricting broadcast traffic to smaller domain.

Virtual Local Area Network (VLAN) enhance security by preventing unauthorized devices from directly accessing sensitive data.

Question 3.1 In your own words, describe the role of DHCP in a network.

DHCP has a number of roles in a Network such as Dynamic ID Assignment which leases unique IP address automatically to new devices connecting into the Network,

Network Parameter Delivery which provides essential parameters like Subnet mask, default gateway and DNS server addresses alongside the IP address

And IP Address Reclamation which reclaims IP address from disconnected devices that left the network back into a pool to be used by other devices, preventing address exhaustion.

Aswell as DHCP Preventing IP Conflicts by tracking all active leases.

Question 3.2 Explain how DHCP improves efficiency compared to manual IP configuration.

DHCP improves efficiency from Manual IP configurations by Automating the IP assignment process instead of manually configuring devices One by One.

It does this by ensuring Centralized Management where you can change your networks DNS server or default gateway only once on DHCP Server. And Automate Record keeping as well as save time with

- DORA processing,
- Error Prevention as result of lacking Human Error.
- And Efficient Address Usage with Lease Management.

Question 3.3 Design a DHCP Scope

DHCP Scope 192.168.100.0/24

Scope Name	Eduvos IT Scope
Network Address	192.168.100.0
Start IP Address	192.168.100.1
End IP Address	192.168.100.254
Subnet Mask	255.255.255.0/24
Default Gateway	192.168.100.1
DNS Server	8.8.8.8
Broadcast Address	192.168.100.255
Lease Duration	8Hrs 0Min 0Sec
Total Usable Host	254

Question 3.4 Explain how a client machine would request and receive an IP address from the DHCP server.

A client machine would Request and Receive an IP address from the DHCP Server using a four-step handshake know as the DORA process.

Which begins with Discover where the client broadcast a message to the limited broadcast IP address across the local subnet to find any available DHCP Server. Discover happens before any offer or request is done.

The process continues with Offer where by DHCP Server receiving a discovery message reserves an available IP address from it pool and offers an IP Address, Subnet mask, Default Gateway, DNS server and lease duration.

The Process continues with Request where the client accepts the offer.

Then Acknowledgment where a chosen DHCP Server finalizes the binding by sending an acknowledgment packet to the client.

- When trying to verify IP assignment in a lab environment students should clear the current IP lease using *ipconfig* / *release* command before requesting a new address, Student should clear any existing configuration.
- Then process by Requestion a New Address (Triggering DORA) using *ipconfig* /*renew* to simulate a new machine booting up or joining a network for the first time.
- Then processed with verifying the detailed leases Parameters using *ipconfig* /*all* to confirm that all intended options were successfully delivered by the server.

Question 3.5 Summarise the benefits Eduvos gains from DHCP automation.

DHCP automation provides Eduvos with a streamlined and zero-touch network experience.

With Seamless Roaming & BYOD support so Students and Staff can easily connect Laptops and Mobile devices on campus without manual configuration.

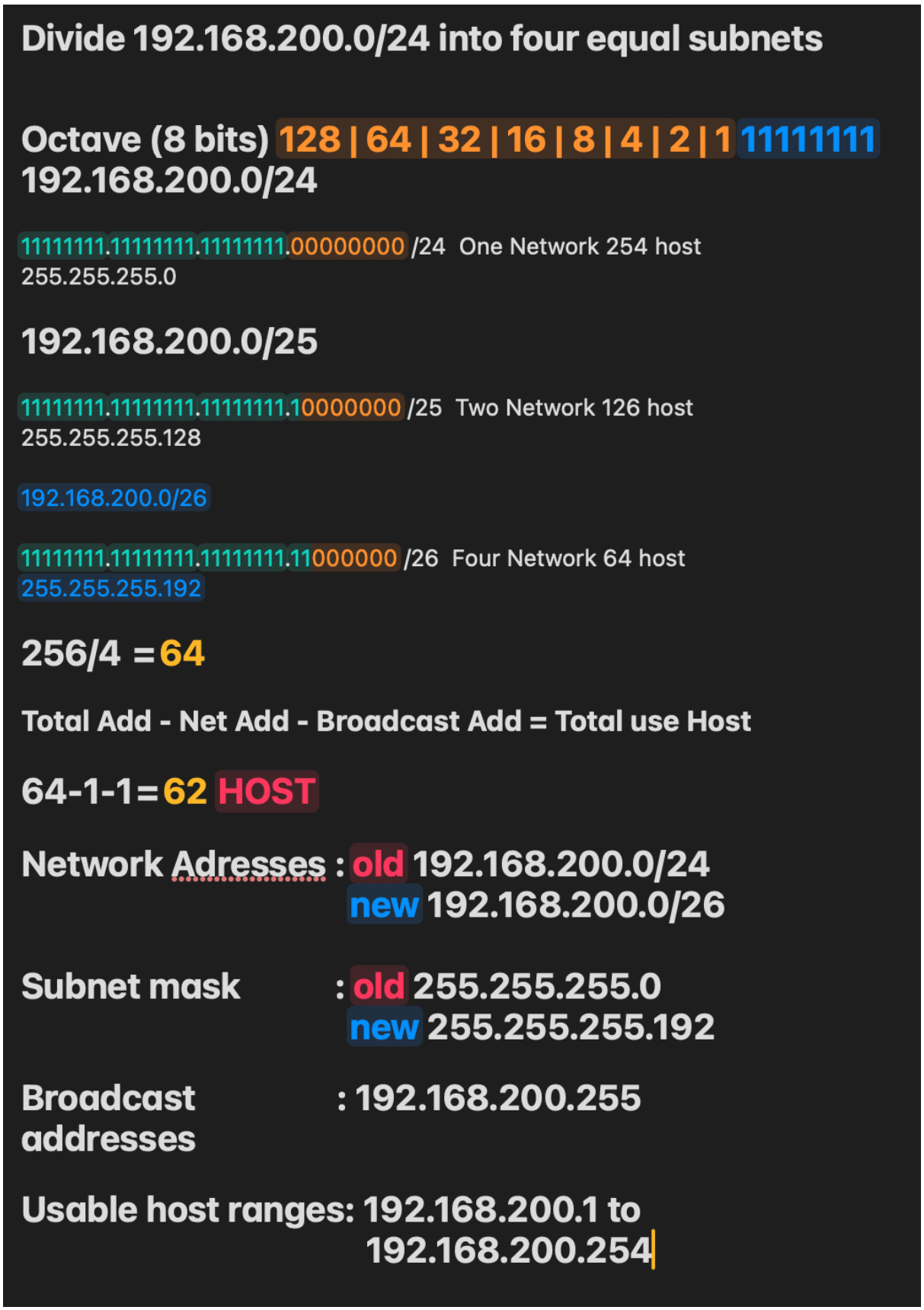
And Zero-Touch IT Administration as IT Staff is freed from manually configuring new devices allowing instant scale toward network.

DHCP also Eliminates Human Error removing mistakes associated with human manual data entry like duplicate IP assignments.

And Optimizes Resource by Recycling and reclaiming IP address

Aswell as Network Security as Automation Systems integrate well with access policies, restricting network access to approved devices.

Question 4.1 Divide the 192.168.200.0/24 network into four equal subnets. You are required to show your calculations step-by-step.



At a CIDR of /26 Network is divided into 4 subnets each with 62 usable hosts. Network borrows 2 bits from Host (128 + 64 = 192) meaning the new subnet is 255.255.255.192. 256/4subnets =64 Host.

Question 4.2 Assign one subnet to each department. You are required to provide a table showing.

Name	Administration	Information System Faculty	Library Services	Students Lab
Subnet	1	2	3	4
Network	192.168.200.0 /26	192.168.200.64 /26	192.168.200.128 /26	192.168.200.192 /26
Usable Host	1 to 62	65 to 126	129 to 190	193 to 254
First Host	192.168.200.1	192.168.200.65	192.168.200.129	192.168.200.193
Last Host	192.168.200.62	192.168.200.126	192.168.200.190	192.168.200.254
Broadcast	192.168.200.63	192.168.200.127	192.168.200.191	192.168.200.255

Question 4.3 Explain how a router (or Layer 3 switch) would enable communication between the four subnets.

A router (or Layer 3 switch) would enable communication between subnets by performing inter-VLAN routing.

Devices in different subnet cannot communicate directly so their traffic is sent to routing devices (default gateway) which reads the destination IP address, Check Routing Table and forward the packet to the correct destination subnet.

Once router or layer 3 switch receives the frame the routing begins as *De-encapsulation* occurs when the devices Strips away the layer 2 Ethernet Header to reveal the layer 3 IP Packet.

Then *Route Lookup* where it examines the destination IP address in the packet header and check it internal Routing Table to find out which interface or network segment the destination subnet belongs to.

Then *Re-encapsulation* when the device wraps the packet in a new Layer 2 Frame using the destinations(or next router) mac address, and sends it out the appropriate interface.

Question 4.4 Describe how you would verify connectivity using tools such as ping or tracert. Provide a simple diagram showing the router connecting the four subnets.

When verifying connections you may use tools such as the Ping command to test basic reachability and latency to a destination. Or tracert/traceroute to map the exact path packets take and identify which specify router or link is dropping the connection.

Ping uses ICMP Echo Request packet to an IP address or hostname and waits for Echo Reply.

Tracert sends packet with increasing Time-to-live(TTL) values, causing each router along the route to reply with an error messages, mapping the path hop-by-hop

Eg. *Ping* 192.168.200.0 or *tracert* 8.8.8.8

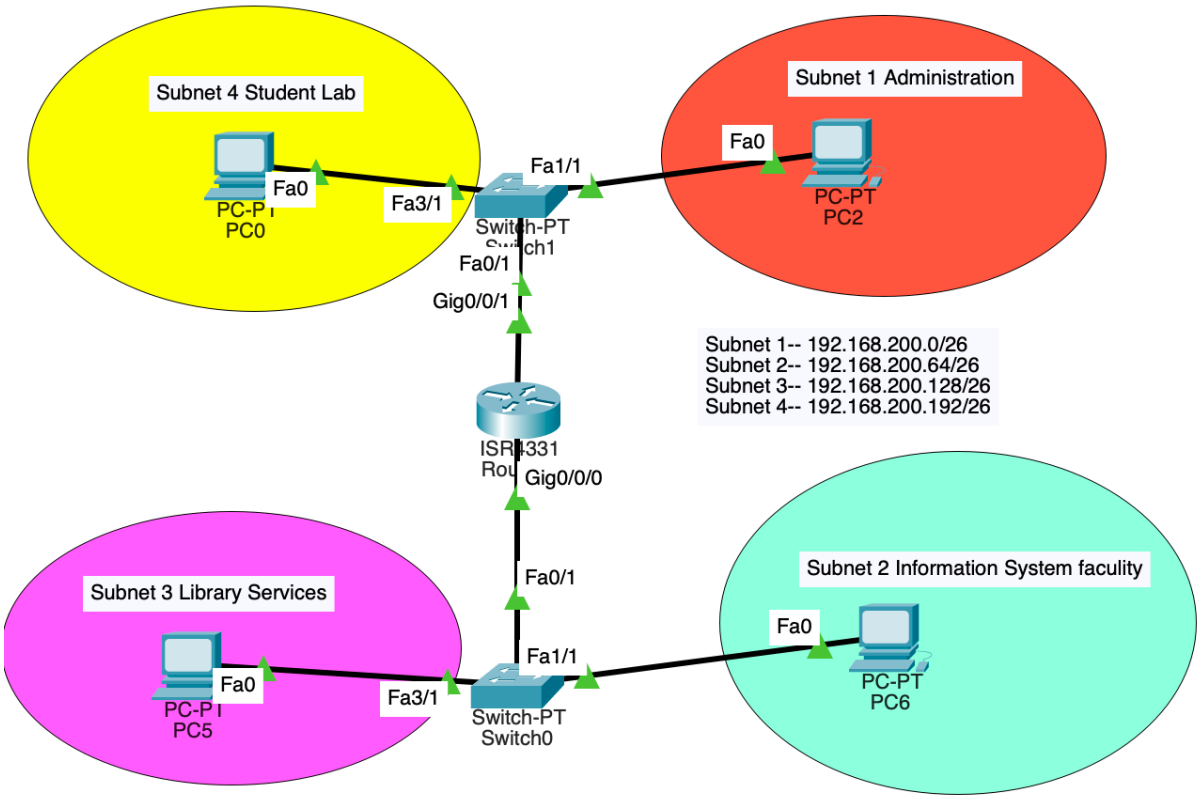


Diagram. Router connecting the four subnets.